Page No:
Date 29

Modified.

* Suggested modifications :-

1) Allow deletion of multiple leaf nodes
2) Display the tree before and after deletion
3) Handle attempts to delete a key that does not exist.

* Problem Statement :-

To write a c++ program to delete multiple specified leaf node from a Binary search Tree (BST), display the tree before and after deletion and handle invalid keys or non leaf nodes.

* Solution :-

The program creates a BST by inserting number It allows the user to enter multiple key for deletion. If key exist and is leaf node, it is deleted. If the key does not exist or is not a leaf node, an appropriate message is displayed. The updated tree is displayed. The updated tree displayed using inorder traversal.

* Objectives :-

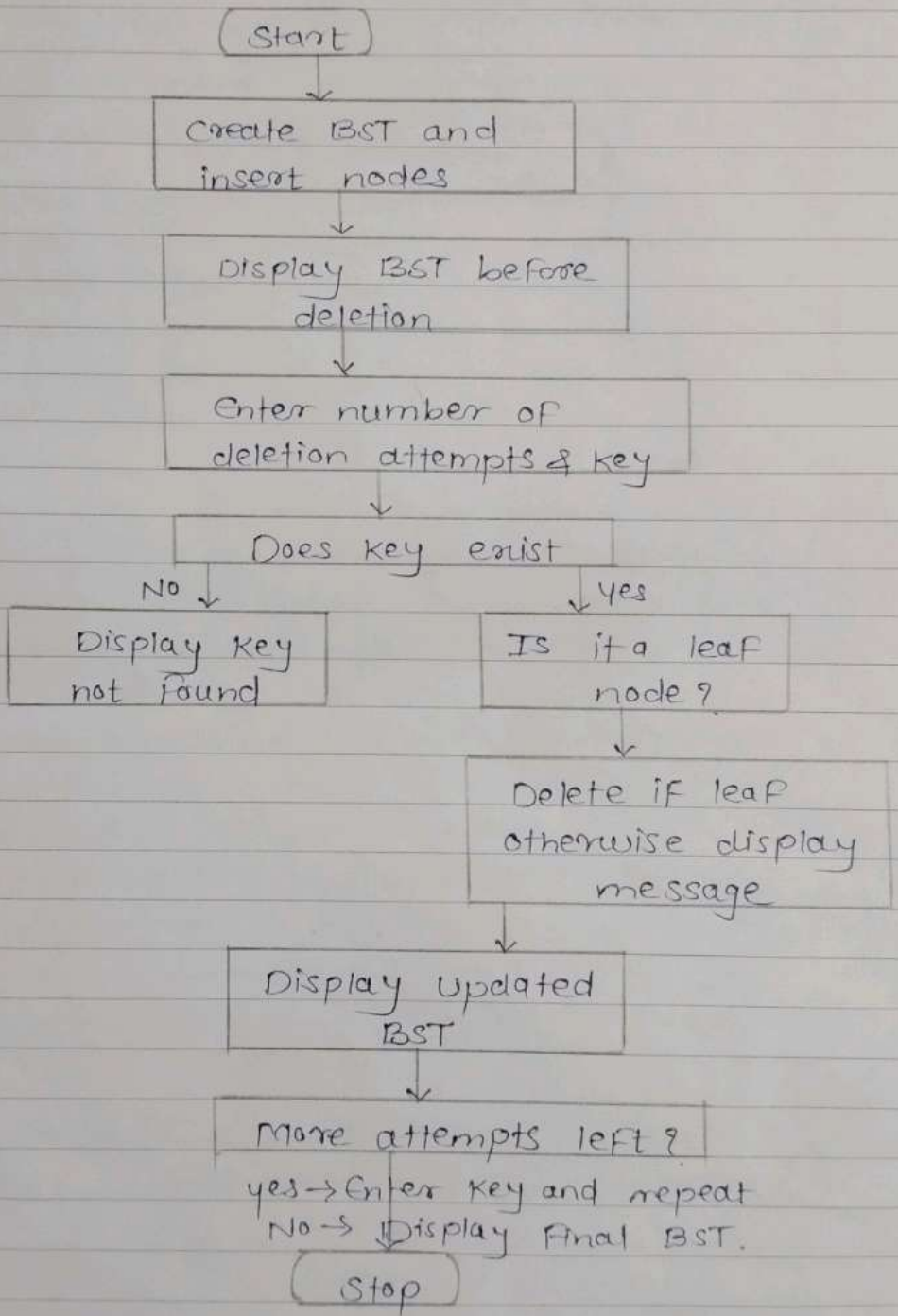
1) To create a Binary search Tree.
2) To delete multiple leaf nodes
3) To display the BST before and after deletion.
4) To handle key that do not exist
5) To maintain the BST property after deletion.

Page No.
Date

* Algorithm :-

1) Start the program.
2) Create an empty BST
3) Enter the number of nodes and their values.
4) Insert the value into the BST.
5) Display the BST using inorder traversal
6) Enter the number of deletion atte
7) Input the key to delete.
8) Search for the specified key.
9) If the does not exist, display an error message.
10) If the key is not leaf node display a message
11) If the key is a leaf node, delete it.
12) Display the update BST after each attempt.
13) Repeat Until all deletion attempts are Completed
14) Display the final BST
15) Stop the program.

* Flowchart :-


Start
Create BST and insert nodes
Display BST before deletion
Enter number of deletion attempts & key
Does key exist
No
Yes
Display key not found
Is it a leaf node ?
Delete if leaf otherwise display message
Display updated BST
More attempts left ?
yes -> Enter key and repeat
No -> Display final BST.
Stop